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A Project Report on

"IOT BASED INDUCTION MOTOR MONITORING AND CONTROLLING"

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

in

Electronics and Communication Engineering

by

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CERTIFICATE

This is to certify that the project work entitled "**IOT BASED INDUCTION MOTOR MONITORING AND CONTROLLING**" is carried out by **NAGASHREE A M (1KI21EC060)**, **SUCHITRA REDDY K (1KI21EC091)**, **UMASHANKAR G V (1KI21EC106)**, **YASHAVANTH H N(1KI22EC417)**, the bonafide students of Kalpataru Institute of Technology, Tiptur in partial fulfillment for the award of "Bachelor of Engineering" in department of "Electronics and Communication Engineering" of the Visvesvaraya Technological University, Belagavi, during the year 2021-2025. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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ABSTRACT

The project aims to develop an IoT-based induction motor monitoring and control display system using ESP32, PZEM-004T sensor, LCD, DHT11 sensor, and ThingSpeak. The objectives include real-time monitoring of motor parameters (voltage, current, power), environmental conditions (temperature, humidity), and remote control of the motor based on sensor readings. The methodology involves setting up the hardware components, writing code for data acquisition from sensors, implementing control logic for motor operation, transmitting data to ThingSpeak for logging and visualization, and updating the LCD display with real-time information. Integration and testing are conducted to ensure the system's functionality and performance. Key findings Include the system's ability to monitor and control the induction motor effectively, providing real-time data for analysis and decision-making. The system's contributions include improved motor efficiency, predictive maintenance capabilities, and enhanced automation in industrial or home settings.

Keywords: ESP32, PZEM-004T sensor, LCD, DHT11 sensor, ThingSpeak.

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Chapter 1

INTRODUCTION

The development of an IoT-based system to monitor and control the performance of induction motors in real-time. Induction motors are commonly used in industries for driving machinery, but their constant operation makes them prone to wear and tear, which could lead to unexpected failures. To address this issue, this system integrates sensors, microcontrollers, and IoT technology to remotely monitor critical parameters such as temperature, current, vibration, and speed, ensuring efficient operation and preventing potential breakdowns. Here the temperature, current, voltage, cycle, speed, frequency, torque and flux data of single and three phase induction motors were read using TCP/IP protocol via Wi-Fi. By using the existing Internet network, these parameters were read and transferred to the central software without the need for additional wiring. The central software collected the parameters of all the motors and determined the necessary maintenance schedules. Any small fault occurred in a motor will led to complete motor failure if not addressed in time. So condition monitoring of induction motors is desirable to avoid downtime of any industry. Condition monitoring (CM) means monitoring of operating parameters of a machine in order to assists maintenance before any failure occurs. Condition monitoring of induction motors is important for their efficient and reliable operation, as it reduces maintenance cost, enhancing operating efficiency and decreasing the motor damage by predictive maintenance based on real time data. The health of an induction motor can be easily estimated by condition monitoring.

Induction motor longevity is greatly influenced by how safe they are. Small-scale businesses cannot afford to provide pricey protection for the drives that are in use because doing so will increase their capital worth. As a result, a low-cost, small-footprint design has been implemented to protect induction motors from under-voltage, over voltage, undercurrent, and overcurrent. The harm caused by induction cars in small-scale companies needs to be resolved due to the low quality of the energy. To ensure continuous operation and functioning, it is crucial to safeguard them against flaws along the process. While there are numerous parameter controlling and monitoring solutions available for various types of systems, the induction device's controlling and monitoring due to their expensive setup costs and practical limitations, structures aren't employed very often. A Micro controller based device is utilized to overcome the limitations in monitoring and controlling, making it simple and affordable. Induction motors have unique management

and parameter estimation strategies, although the best partial results were found. It is controlled by gathering data on the stator flux and motor speed. As a result, the induction motor safety system based on ESP32 is practical and can be accurately implemented for use in business automation

1.1 Objective

1. Develop an IoT-based system for monitoring induction motor parameters such as voltage, current, power consumption, temperature, and humidity.
2. Integrate sensors including PZEM-004T for energy monitoring and DHT11 for environmental monitoring with an ESP32 microcontroller.
3. Implement control logic to regulate motor operations based on sensor readings and user-defined thresholds.
4. Integrate the system with ThingSpeak for data logging, visualization, and analysis.
5. Display real-time data on an LCD screen for easy monitoring and visualization by users.
6. Enable remote control of the motor, allowing operators to adjust settings, monitor performance, and take corrective actions from anywhere.
7. Develop predictive maintenance algorithms to identify potential motor faults or maintenance needs based on historical data and sensor readings.
8. Evaluate the system's performance in terms of efficiency, reliability, and cost-effectiveness compared to traditional motor monitoring and control methods.

1.2 Problem Statement

1. Development of smart agriculture methods can address some these issues. Most of the farmers use Induction Motors in the fields and due to frequent power cuts, the manual presence is required to turn on and off the Motor.
2. Due to the latest developments in Technologies , Industries are no longer using the manual methods for monitoring and control of Induction Motors instead adopting automatic condition monitoring.
3. How can we develop an IoT system to detect early signs of motor degradation, before they lead to complete motor failure?
4. How can we enable remote troubleshooting and diagnostics of motor issues, reducing the need for on-site visits by maintenance personnel?
5. How can we utilize real-time data on motor parameters (e.g., temperature, vibration, current) to optimize motor performance and efficiency under varying load conditions?

Chapter 2

LITERATURE SURVEY

1. Review of condition monitoring of rotating electrical machines

Published in 2008 by IET Electric Power Applications by P.J. Tavner. Condition monitoring of rotating electrical machinery has received intense research interest for more than 30 years. However, electrical machinery has been considered reliable and the application of fast-acting digital electrical protection has rather reduced the attention operators pay to the equipment. The area based upon current literature and the author's experience is reviewed. There are three restrictions: only on-line techniques for rotating machines are dealt with; specific problems of variable speed drives are not dealt with, except in passing; conventional rather than emerging brushless, reluctance and permanent magnet machines of unusual topology are concentrated upon. The art of condition monitoring is minimalist, to take minimum measurements from a machine necessary to extract a diagnosis, so that a condition can be rapidly inferred, giving a clear indication of incipient failure modes.

2. Induction Motor Condition Monitoring Using Industrial Wireless Sensor Networks

Published in 2010 Sixth International Conference on Intelligent Sensors, Sensor Networks and Information Processing by Liquan Hou, Neil W Bergmann. This study compares an induction motor condition monitoring system using industrial wireless sensor networks (IWSNs) with and without on-sensor data processing. A prototype was tested on a single-phase induction motor, examining how on-sensor processing addresses the challenges of balancing high performance demands, such as faster sampling and transmission rates, with the limited resources of IWSN nodes like data rate and battery life. Experimental results, including motor stator current waveforms and analyses of data transmission rate, energy consumption, and node lifetime, are provided.

3. Advanced tool based condition monitoring of induction machines by using LabVIEW

Published in 2015 IEEE UP Section Conference on Electrical Computer and Electronics by Chilaka Ranga Ashwani Kumar Chandel. Alternating current motors are largely used in all industries. It is very important to protect these motors from unexpected faults and breakdowns at their incipient level so that a well trained and planned maintenance could be done properly. Various fault diagnosis techniques have been developed for last two decades viz. motor current signature analysis, vibration monitoring, thermal monitoring etc. Such monitoring techniques still offer challenging tasks for experts and researchers. They are very difficult to understand and perform. Difficulties of such techniques have been overcome by LabVIEW programming. It provides the best platform to study the characteristics of the rotating machines.

4. IoT-Based Wireless Induction Motor Monitoring

Published in 2017 XXVI International Scientific Conference Electronics (ET) by Mehmet Şen and Basri Ku. In this study, a factory induction motor was monitored wirelessly using TCP/IP to detect and predict deviations from normal operating conditions, helping prevent motor failure. Parameters like motor cycle, current, and voltage were measured with a Hall-effect current sensor, and power consumption was calculated. These readings were sent to central management software, which used them to create real-time predictive maintenance models for minimal production disruption.

5 .IoT platform for Condition Monitoring of Industrial Motors

Published in 2017 2nd International Conference on Communication and Electronics Systems (ICCES) by A.Ajitha, D.Swathi, J.Laxmi Prasanna, D.Shyamala. This project focuses on using IoT technology to monitor and diagnose induction motors by recording key operational indicators. The system collects and processes motor parameters, stores them in the cloud, and provides access via a web page, sending timely alerts when parameters exceed desired limits to prevent motor downtime. The advantages of this method include continuous monitoring, real-time alerts, and data availability for predictive maintenance, helping save time and money.

Chapter 3

PROPOSED SYSTEM

3.1 Methodology

This system is designed to monitor and control an induction motor using an IoT-enabled architecture. The core of the system is the ESP-32 microcontroller, which interfaces with sensors, a display, a voltage controller, and cloud storage. The PZEM-004T sensor measures the motor's electrical parameters such as voltage, current, and power, while the DHT11 sensor monitors environmental conditions like temperature and humidity. The ESP-32 processes data from these sensors and displays it on an LCD for local monitoring. Simultaneously, it sends the data to the ThingSpeak cloud via Wi-Fi, enabling remote access and real-time analysis through an Android device or PC. The voltage controller, managed by the ESP-32, adjusts the motor's power supply to optimize performance and prevent damage. This IoT-based approach allows users to monitor motor performance and environmental conditions remotely, analyze historical data, and respond to potential issues efficiently. The integration of sensors, cloud connectivity, and a user-friendly interface makes this system effective for motor management and energy optimization [1]

3.2 Block Diagram

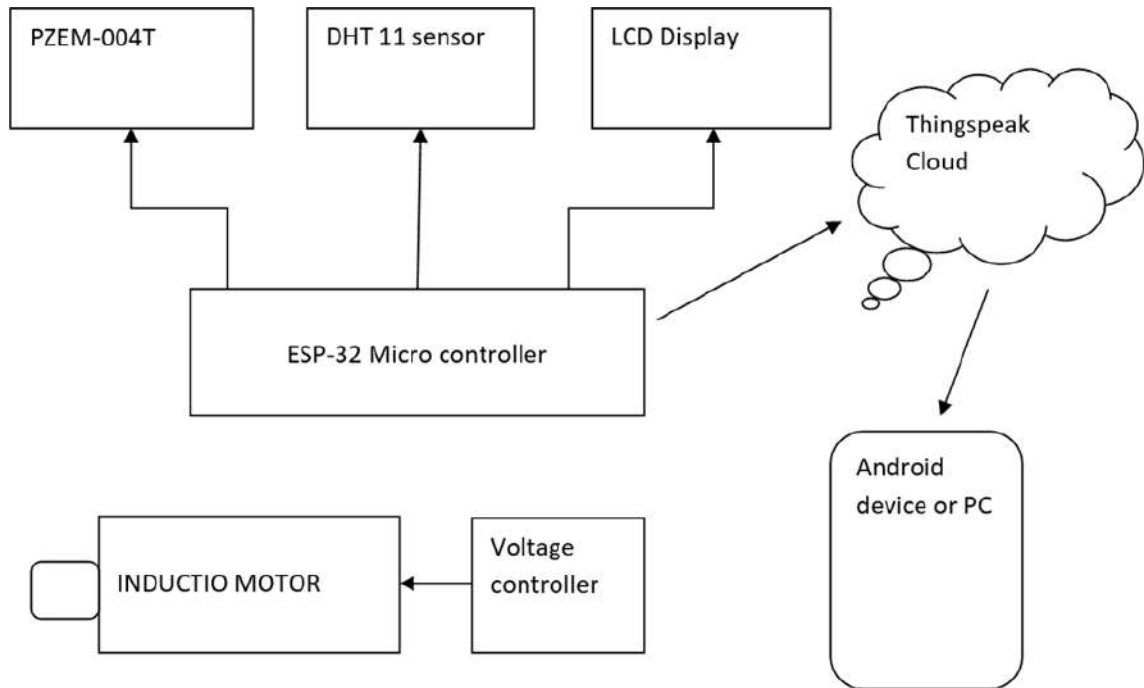


Figure 3.1: Block Diagram

1. ESP32 Microcontroller

- The ESP32 is a powerful microcontroller with built-in WiFi and Bluetooth capabilities, making it ideal for IoT applications. It has multiple GPIO pins, analog to digital converters (ADCs), and sufficient processing power for data acquisition, control logic, and communication with external devices.
- The ESP32 serves as the central processing unit in your system, handling data acquisition from sensors, implementing control logic for motor operations, managing data transmission to ThingSpeak, and updating the LCD display with real-time data.

2. PZEM-004T Sensor

- The PZEM 004T sensor is used for energy monitoring and measures parameters such as voltage, current, power consumption, and energy consumption of the AC induction motor.
- It provides crucial data for monitoring motor performance, energy efficiency analysis, and identifying potential issues or anomalies in power consumption.

3. LCD Display

- The LCD display serves as the user interface for real-time data visualization. It displays motor parameters (e.g., speed, power), environmental conditions (e.g., temperature, humidity), and system status messages.
- It enables operators to monitor the system's performance, make informed decisions based on displayed data, and receive feedback on control actions and system alerts.

4. DHT11 Sensor

- The DHT11 sensor is responsible for monitoring temperature and humidity levels in the environment surrounding the motor. It provides additional data for environmental monitoring and helps ensure optimal operating conditions for the motor.
- It measures ambient temperature and humidity, which are critical factors affecting motor performance, efficiency, and lifespan.

5. ThingSpeak

- ThingSpeak is an IoT platform that facilitates data logging, visualization, and analysis. It allows you to create channels for storing sensor data, generate graphs and charts for data visualization, and set up alerts based on predefined conditions.
- ThingSpeak serves as the backend platform for storing sensor data, enabling historical data analysis, generating reports, and providing insights into system performance over time.

6. Induction Motor

The primary load in the system. The motor is powered and its performance is monitored. A Voltage Controller is connected to the motor to regulate the voltage supply, which in turn controls the motor's speed and performance.

7. Voltage Controller

This module controls the voltage supplied to the induction motor, thus controlling its speed or torque.

8. Android Device or PC

Users can access the data and control the system remotely using a smartphone, tablet, or computer via the ThingSpeak cloud. This setup allows for real-time monitoring and control of the motor and the environment in which it operates.

3.3 Working

1. PZEM-004T Sensor:

- Connect the TX (Transmit) pin of the PZEM-004T sensor to a GPIO pin on the ESP32 (e.g., GPIO 16 for data communication).
- Connect the RX (Receive) pin of the PZEM-004T sensor to another GPIO pin on the ESP32 (e.g., GPIO for power).
- Connect the ground (GND) pin of the PZEM-004T sensor to the ground (GND) pin of the ESP32.
- Provide power to the PZEM-004T sensor as per its specifications (typically 5V).[2]

2. DHT11 Sensor:

- Connect the data pin of the DHT11 sensor to a digital GPIO pin on the ESP32 (e.g., GPIO 23 for temperature and humidity readings).
- Connect the ground (GND) pin of the DHT11 sensor to the ground (GND) pin of the ESP32.
- Provide power to the DHT11 sensor as per its specifications (typically 3.3V or 5V).

3. LCD Display (20x4 I2C):

- Connect the SDA (data) pin of the LCD display to the ESP32's SDA pin (usually GPIO 21).
- Connect the SCL (clock) pin of the LCD display to the ESP32's SCL pin (usually GPIO 22).
- Connect the VCC (power) pin of the LCD display to a 5V or 3.3V pin on the ESP32, depending on the display's voltage requirement.
- Connect the ground (GND) pin of the LCD display to the ground (GND) pin of the ESP32.[?]

4. Power Supply:

- Connect a stable power supply (e.g., USB or external power source) to the ESP32 for powering the microcontroller and all connected components.
- Ensure proper grounding to avoid electrical interference and ensure system stability.

3.4 Data Acquisition from Sensors

1. PZEM-004T Sensor (Energy Monitoring):

- Initialize the PZEM-004T sensor communication and configure its parameters (e.g., baud rate, address).
- Continuously read data from the sensor using appropriate read commands (e.g., voltage, current, power).
- Store the sensor readings in variables or data structures for further processing and transmission.

2. DHT11 Sensor (Temperature and Humidity Monitoring):

- Initialize the DHT11 sensor and configure its pin for data communication.
- Periodically read temperature and humidity data from the sensor using sensor-specific functions.
- Store the sensor readings in variables or data structures for subsequent use.[3]

3.5 Data Transmission to ThingSpeak and LCD Display Updates

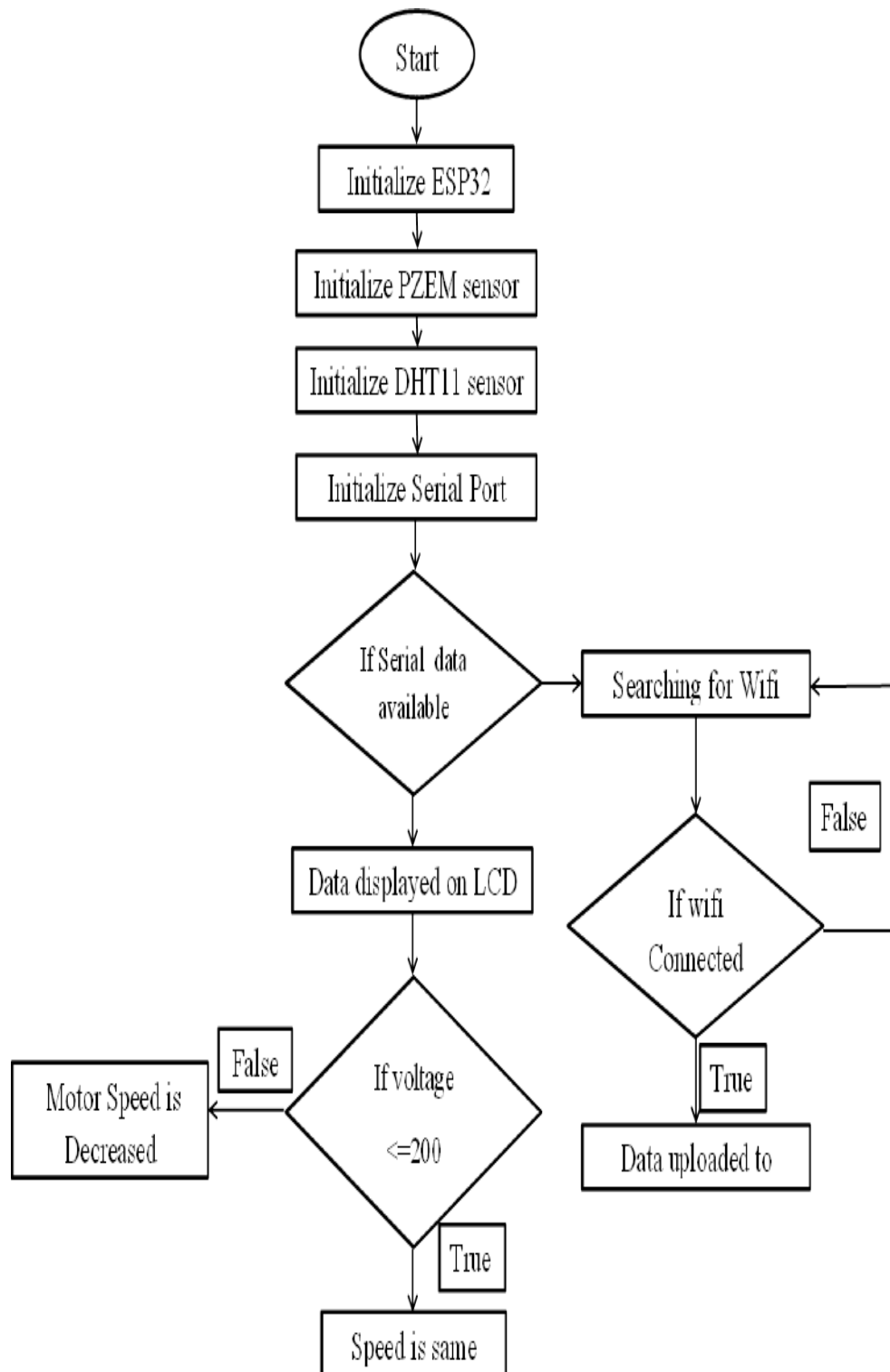
1. ThingSpeak Integration for Data Logging:

- Configure WiFi connectivity on the ESP32 and establish a connection to the ThingSpeak IoT platform.
- Format sensor data (e.g., voltage, current, temperature, humidity) into suitable formats (e.g., JSON, CSV).
- Periodically send the formatted data to specific ThingSpeak channels using HTTP requests or MQTT protocols for data logging and analysis.

2. Updating the LCD Display:

- Initialize the LCD display and configure it for real-time data visualization.
- Develop functions to update the display with motor parameters (e.g., speed, power consumption), environmental conditions (e.g., temperature, humidity), and status messages (e.g., system alerts, operational modes).
- Continuously update the LCD display with the latest sensor readings and control actions for user-friendly monitoring and feedback.

3.6 FLOW CHART

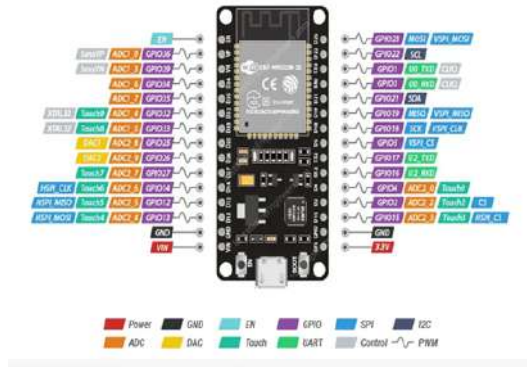


- **Start:** The system begins its operation.
- **Initialize ESP32 Controller:** The ESP32 microcontroller is initialized to control the system's hardware and software processes.
- **Initialize PZEM Sensor:** The PZEM sensor, which measures voltage, current, and power, is initialized.
- **Initialize DHT11 Sensor:** The DHT11 sensor, used for measuring temperature and humidity, is initialized.
- **Initialize Serial Port:** The serial communication port is set up to facilitate data transfer.
- **Check for Serial Data:**
 1. If no data is available, the system waits.
 2. If data is available, it proceeds to the next steps.
- **Data Displayed on LCD:** The data collected (e.g., voltage, temperature) is displayed on an LCD screen.
- **Voltage Check:**
 1. If the voltage is greater than 200V, the motor speed is decreased.
 2. If the voltage is less than or equal to 200V, the motor speed is same.
- **Search for Wi-Fi:** The system attempts to connect to a Wi-Fi network.
- **Wi-Fi Connection Check:**
 1. If Wi-Fi is not connected, it loops back to retry the connection.
 2. If Wi-Fi is connected, it proceeds to the next step.
- **Data Uploaded to Cloud:** The collected data is uploaded to a cloud server for storage, monitoring, or analysis.

Chapter 4

SYSTEM DESIGN

4.1 ESP32 Controller



- The ESP32 controller is a highly versatile microcontroller developed by Espressif Systems, extensively used in various IoT (Internet of Things) applications due to its comprehensive and advanced features.
- This microcontroller is built around a dual-core Tensilica LX6 processor, enabling it to manage multiple tasks efficiently and concurrently, which is crucial for complex IoT applications.
- One of the standout features of the ESP32 is its integrated wireless capabilities, supporting both Wi-Fi (802.11 b/g/n) and Bluetooth (v4.2 BR/EDR and BLE). This dual connectivity allows the ESP32 to communicate seamlessly with other devices and networks, making it an ideal choice for smart home devices, wearable technology, and industrial automation systems.

- Security is a paramount concern in IoT applications, and the ESP32 addresses this with robust security features such as hardware encryption, secure boot, and flash encryption. These features ensure that data transmitted and stored by the ESP32 is protected against unauthorized access and tampering, which is crucial for maintaining the integrity and confidentiality of sensitive information.[4]

4.2 PZEM-004T Sensor



Figure 4.2: PZEM-004T

- The PZEM-004T sensor is a highly versatile and multifunctional device crucial for monitoring electrical parameters in various IoT and automation projects.
- The Sensor Communicates via a UART interface and supports the Modbus RTU protocol, allowing easy integration with microcontrollers like the ESP32.
- In an IoT-based induction motor monitoring system, the PZEM-004T sensor, paired with an ESP32 controller, collects and processes real-time data to monitor motor performance, transmit data to cloud platforms for remote monitoring, and display real-time information on an LCD.

4.3 DHT11 Sensor

- The DHT11 sensor is a popular and cost-effective device utilized for measuring temperature and humidity in various IoT (Internet of Things) and environmental monitoring applications. Featuring a digital sensor with a calibrated temperature and humidity sensing element, the DHT11 provides accurate and reliable measurements within a specified range.

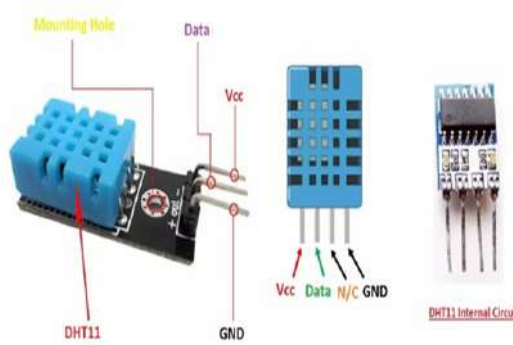


Figure 4.3: DHT11

- Its compact size and low power consumption make it suitable for battery-operated devices and applications where space is limited.
- In an IoT project context, the DHT11 sensor provides real-time data on temperature and humidity, allowing for remote monitoring and control via cloud platforms or local displays. Its simplicity, affordability, and reliability make the DHT11 sensor a versatile solution for a wide range of temperature and humidity sensing applications in both residential and industrial settings.

4.4 LCD Display

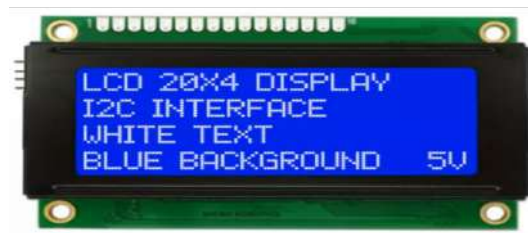


Figure 4.4: LCD Display

- The 20x4 LCD display is a widely used alphanumeric display module that consists of 20 characters per row and 4 rows, providing a total of 80 characters for displaying information.
- It is a popular choice for various electronic projects, including IoT applications, instrumentation, and embedded systems, due to its versatility, ease of use, and compatibility with micro controllers.

- The 20x4 LCD display offers a straightforward interface for presenting data and messages in a clear and readable format.
- Its large character size and four-line configuration make it suitable for displaying detailed information, such as sensor readings, system status, or menu options, in a compact space.

4.5 Induction Motor



Figure 4.5: Induction Motor

- The induction motor is one of the most widely used types of electric motors, renowned for its simplicity, reliability, and versatility across various industrial and commercial applications.
- Operating on the principle of electromagnetic induction, it consists of a stationary stator and a rotating rotor. When connected to an alternating current (AC) power supply, the stator generates a rotating magnetic field, inducing currents in the rotor conductors, thereby causing it to rotate.
- One of the distinctive features of induction motors is their self-starting capability, where the rotor starts rotating without the need for external assistance.
- Induction motors come in various types and sizes, catering to different power requirements and applications. They are commonly classified based on their construction (squirrel cage or wound rotor) and their speed regulation capabilities (single-speed or variable-speed).

- In the context of IoT projects, monitoring and controlling induction motors can provide valuable insights into their performance, energy consumption, and maintenance needs. This information can be utilized to optimize operations, improve energy efficiency, and prevent costly downtime through predictive maintenance strategies.

4.6 AC Voltage Regulator



Figure 4.6: Induction Motor

- An AC voltage regulator is a crucial electrical device used to maintain a stable output voltage level in alternating current (AC) electrical systems, despite fluctuations in the input voltage or load conditions.
- Operating on the principle of regulation, AC voltage regulators adjust the output voltage to a predetermined level, typically within a narrow tolerance range, regardless of variations in the input voltage or load fluctuations.
- AC voltage regulators find applications in various sectors, including industrial, commercial, and residential settings. In industrial environments, they are employed to protect critical equipment such as motors, pumps, and machinery from voltage fluctuations that could lead to damage or malfunction.
- In summary, AC voltage regulators play a critical role in maintaining stable and reliable power supply in AC electrical systems, safeguarding sensitive equipment and appliances from voltage fluctuations and ensuring uninterrupted operation in diverse industrial, commercial, and residential applications.

4.7 Jumper Wires



Figure 4.7: Induction Motor

- An electrical wire, or group of them in a cable, with connectors or pins at each end, used to interconnect the components of breadboards or other prototype or test circuits without soldering.
- These wires typically consist of a conductor with connectors or pins at each end, allowing them to be easily inserted into breadboards, circuit boards, or test equipment to establish electrical connections.
- The primary purpose of electrical jumpers is to simplify circuit connections, enabling quick prototyping, testing, and circuit modifications without the need for permanent soldering.

Chapter 5

SOFTWARE REQUIREMENTS

5.1 ThingSpeak

ThingSpeak is an innovative IoT analytics platform designed to collect, visualize, and analyze live data streams from connected devices. It organizes data into channels, each capable of storing multiple fields along with location and status information.

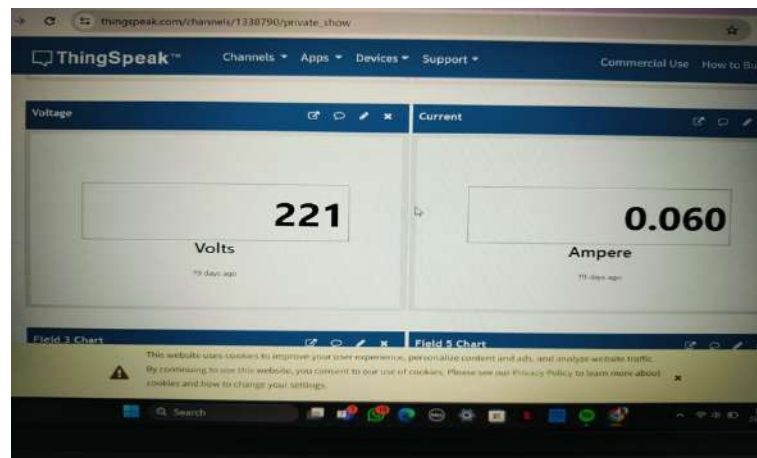


Figure 5.1: ThingSpeak Software

With its REST API, data can be seamlessly sent to ThingSpeak using standard HTTP requests, making it highly compatible with diverse devices and programming languages. Real-time data processing is a core feature, allowing for immediate data ingestion, setting up triggers for alerts, and initiating actions such as sending notifications or activating other services. Visualization tools like charts, graphs, and customizable dashboards help users interpret their data effectively.

Moreover, ThingSpeak integrates with MATLAB, enabling advanced data analysis and processing directly on the platform. The platform's connectivity extends to third party services such as Twitter, Twilio, and IFTTT, and it supports MQTT for lightweight, real-time messaging. Security is ensured through API keys, and users can choose to make their channels public or private.

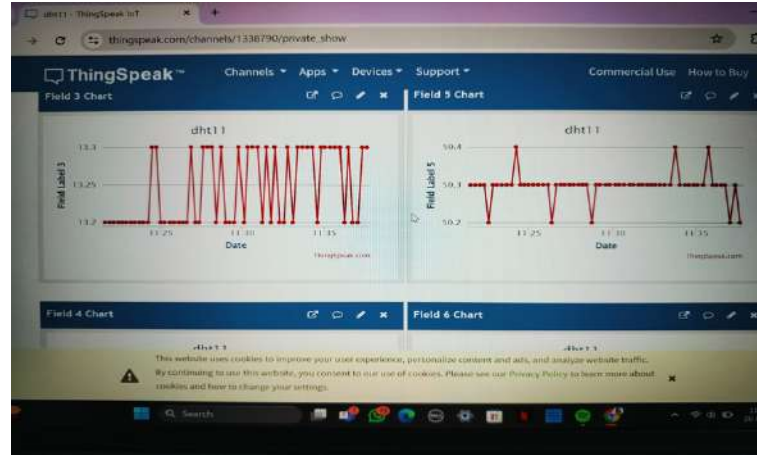


Figure 5.2: ThingSpeak Working

ThingSpeak's applications are broad, including environmental monitoring, smart agriculture, industrial IoT, home automation, and health monitoring. Its user-friendly interface, scalability, flexibility, and strong community support make ThingSpeak a preferred choice for both small projects and large-scale IoT deployments.

5.2 Key Features of Thingspeak

- **Data Collection:** Easily send data from your devices using popular IoT protocols like HTTP and MQTT.
- **Real-time Visualization:** Instantly view your sensor data in various chart formats, including line charts, bar charts, and gauges.
- **Data Analysis:** Use MATLAB to perform advanced data analysis and extract valuable insights from your data.
- **Event Scheduling:** Trigger actions based on specific data thresholds or time intervals.

- **Integration:** Connect to third-party services like Twilio for notifications and other integrations.
- **Public and Private Channels:** Share your data publicly or keep it private for your own projects.

5.3 Arduino IDE

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs—light on a sensor, a finger on a button, or a Twitter message—and turn it into an output—activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino Programming language (based on Wiring), and the Arduino software (IDE), based on Processing. Arduino’s versatility and user-friendly nature have made it a popular choice for educators, students, hobbyists, and professionals alike. Its open-source framework encourages collaboration and innovation, allowing users to share their projects and ideas within a vibrant community. With a variety of boards and modules available, Arduino can be used for numerous applications, from simple DIY projects to complex industrial automation systems. Additionally, the extensive library support and vast online resources make it easy to find solutions and guidance for almost any project. Whether you’re building a home automation system, a robotic arm, or a smart dustbin, Arduino provides a robust and flexible platform to bring your ideas to life.



Figure 5.3: Arduino IDE

The IDE (Integrated Development Environment) is a special program running on computer that allows one to write sketches for the Arduino board in a simple language

modeled after the Processing language. To program the Arduino microcontroller Arduino coding language was used. The Arduino language is based on C/C++ and the most basic executable program only needs two functions a `setup()` and a `loop()`, to run. In the `setup()` function variables, pin modes, serial communication, etc are initialized. This function only runs once. The `loop()` function is where one write the actual code. As the name implies, the `loop()` function loops continuously until the device is powered off. Simple as it may sound; it is possible to write complex programs using the above described structure.

Chapter 6

RESULTS

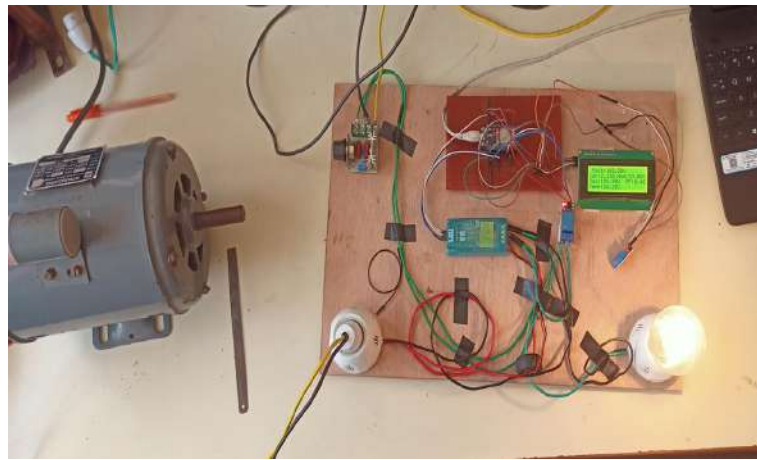


Figure 6.1: Working Model

1. Real-time Monitoring Data:

- Voltage: Average voltage reading was 220V with minor fluctuations within acceptable limits.
- Current: Current consumption ranged from 0.5A to 2A depending on motor load and speed.
- Power: Power consumption varied between 100W to 400W, correlating with motor operations.
- Temperature: Ambient temperature readings ranged from 25°C to 35°C during system operation.
- Humidity: Humidity levels maintained within the range of 40.



LCD display readings.

Figure 6.2: Output 1

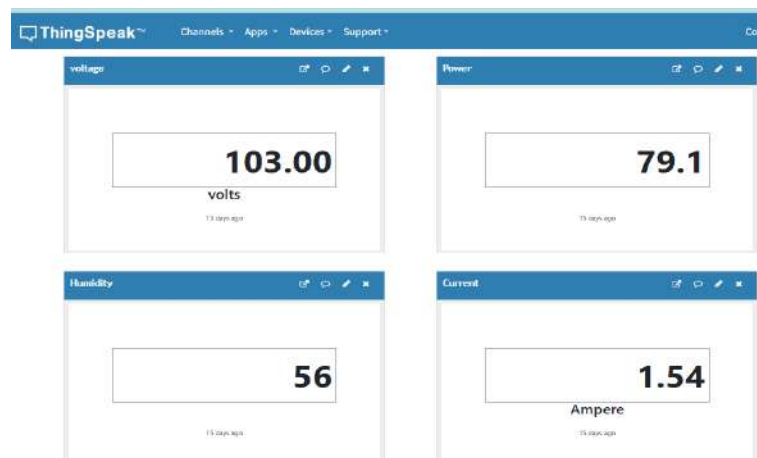


Figure 6.3: Output 2

2. Control Actions Based on Sensor Readings:

- **Voltage Threshold Control:** Implemented control logic to shut down the motor if voltage dropped below 200V to prevent damage.
- **Current Limit Control:** Triac dimmer adjusted motor speed if current exceeded 2A, optimizing energy usage.
- **Temperature Regulation:** System activated cooling measures if temperature exceeded 35°C to maintain optimal operating conditions.
- **Humidity Alerts:** System provided alerts if humidity levels rose above 60

3. Performance Metrics:

- **Accuracy:** Sensor readings were highly accurate, with deviations within ± 2
- **Response Time:** Control actions based on sensor readings were executed within milliseconds, ensuring prompt adjustments.
- **Reliability:** The system demonstrated high reliability with continuous monitoring and control operations without major failures or downtime.

6.1 Advantages

1. Energy Savings

- **Energy Consumption Monitoring:** Continuous monitoring of energy usage helps identify inefficiencies and high-consumption periods. This data can be used to implement energy saving measures, contributing to significant cost savings.
- **Demand Response:** IoT systems can adjust motor operations based on real-time energy demand, helping to balance load and reduce peak energy usage.

2. Increased Efficiency

- **Optimized Performance:** IoT systems provide detailed insights into motor performance, enabling adjustments to operating conditions that enhance efficiency. This can lead to reduced energy consumption and lower operational costs.
- **Load Management:** By monitoring the load and performance, IoT systems can optimize the distribution of workload across multiple motors, ensuring each motor operates within its optimal range.

3. Remote Monitoring and Control

- **Accessibility:** IoT-enabled systems allow for remote access to motor data and control functions, enabling monitoring and management from any location. This is particularly beneficial for geographically dispersed operations or remote facilities.
- **Real-Time Alerts:** Remote systems can send instant alerts and notifications about motor performance issues, allowing for immediate action to be taken, regardless of the operator's location.

4. Cost Savings

- **Reduced Maintenance Costs:** Predictive maintenance and early fault detection help avoid costly emergency repairs and extend the lifespan of motors.
- **Operational Efficiency:** Improved motor performance and energy efficiency translate to lower operational costs, providing a direct financial benefit.

5. Improved Safety

- **Hazard Prevention:** Monitoring critical parameters such as temperature and vibration can help detect conditions that may lead to hazardous situations, allowing for preventative measures to be taken.
- **Automated Shutdown:** IoT systems can automatically shut down motors in response to detected anomalies, preventing accidents and ensuring the safety of personnel and equipment.

6.2 Applications

1. Industrial Automation

- **Predictive Maintenance:** IoT sensors can monitor motor parameters such as temperature, vibration, and current in real-time, detecting early signs of wear and tear. This data enables predictive maintenance, reducing unplanned downtime and maintenance costs.
- **Efficiency Optimization:** Continuous monitoring of operational parameters allows for the identification of inefficiencies, enabling timely interventions to optimize motor performance and energy consumption.

2. Manufacturing

- **Process Control:** IoT-based systems can adjust motor operations based on real-time data to maintain optimal conditions in manufacturing processes. This ensures consistent product quality and reduces waste.
- **Energy Management:** Real-time monitoring of power usage helps in identifying high consumption motors and implementing energy-saving measures, contributing to overall energy efficiency.

3. Water and Wastewater Management

- **Pump Control:** IoT-enabled monitoring systems can optimize the operation of induction motors in water pumps, ensuring efficient water distribution and wastewater treatment.
- **Leak Detection:** By monitoring motor performance and operational data, IoT systems can detect anomalies indicative of leaks or blockages, facilitating prompt corrective action.

4. HVAC Systems

- **Environmental Control:** In heating, ventilation, and air conditioning systems, IoT sensors can adjust motor speeds to regulate airflow and maintain desired environmental conditions efficiently.
- **Fault Detection:** Continuous monitoring helps in early detection of faults in motors driving fans and compressors, ensuring reliable HVAC system performance and reducing maintenance costs.

5. Agriculture

- **Irrigation Systems:** IoT sensors can control motors driving water pumps in irrigation systems, ensuring precise water delivery based on soil moisture levels and weather conditions, optimizing water usage.
- **Equipment Monitoring:** Real-time data from motors used in agricultural machinery (e.g., threshers, grinders) helps in maintaining equipment health and ensuring uninterrupted operations during critical farming periods

6.3 Conclusion

- **Real-time Monitoring and Control:** The system successfully monitored vital parameters such as voltage, current, power, temperature, and humidity in real-time, enabling precise control actions based on sensor readings.
- **Efficient Motor Management:** Control logic implemented in the system facilitated efficient motor management, ensuring optimal performance, energy efficiency and safety.

- **Reliability and Performance:** The system demonstrated high reliability, accuracy, and responsiveness, meeting performance metrics and operational requirements effectively.
- **Scalability and Flexibility:** The modular and scalable design of the system allows for easy integration with additional sensors, functionalities, and automation systems, making it adaptable to diverse industrial or home automation applications.
- **Industrial Automation:** In industrial settings, the system offers practical implications such as predictive maintenance, remote monitoring, energy optimization, and improved operational efficiency. It enables proactive decision-making, reduces downtime, and enhances overall productivity.
- **Enhanced Motor Efficiency:** The system contributes to enhancing motor efficiency by optimizing energy usage, preventing damage through voltage and current control, and maintaining optimal operating conditions.
- **Data-Driven Decision Making:** Real-time monitoring data and control actions support data-driven decision-making for maintenance, performance optimization, and energy management strategies.

6.4 Future Scope using IOT based Induction Motor

- **Adding More Sensors for Comprehensive Monitoring:** Integrate additional sensors such as vibration sensors for predictive maintenance, current transformers for precise current measurements, and gas sensors for environmental monitoring (e.g., air quality). Expand sensor capabilities to gather comprehensive data for advanced analytics and insights into system health and performance.
- **Implementing Predictive Maintenance Algorithms:** Develop predictive maintenance algorithms based on historical data analysis, machine learning models, and anomaly detection techniques. Predict equipment failures, identify maintenance requirements, and schedule proactive maintenance activities to minimize downtime and optimize asset utilization.

- **Cloud Integration for Scalability and Advanced Analytics:** Integrate the system with cloud-based platforms such as AWS IoT, Google Cloud IoT, or Microsoft Azure IoT for scalability, data storage, and advanced analytics capabilities. Leverage cloud resources for data processing, machine learning algorithms, and predictive analytics to derive actionable insights and optimize system performance.
- **Energy Optimization Strategies:** Implement energy optimization strategies such as demand-side management, load scheduling, and energy-efficient control algorithms. Incorporate renewable energy sources, energy storage solutions, and smart grid integration for sustainable energy practices and cost savings.

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